GLIBENCLAMIDE

Prolonged hypoglycaemia

Avoid in elderly

CHEMICAL NAME 5-chloro-N-[2-[4-(cyclohexylcarbamoylsulfamoyl)phenyl]ethyl]-2-methoxybenzamide (glyburide)

CLASS Second-generation sulfonylurea (long-acting)

TRADE NAMES Daonil · Euglucon · Glucovance (with metformin) · Diabeta

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes (when metformin insufficient or not tolerated).
- Gestational diabetes (some guidelines).
- Neonatal diabetes due to KATP channel mutations (specialist).

ADMINISTRATION

- **PO** with breakfast (or first main meal); higher doses split.
- Do not skip meals after taking.

DOSAGES

- **Start:** 2.5–5 mg daily with breakfast.
- **Maintenance:** 1.25–20 mg/day (max 20 mg/day, non-micronised; micronised max 12 mg), single or divided doses; increase by ≤2.5 mg weekly.
- Avoid in elderly and renal impairment (prefer gliclazide).

PHARMACOKINETICS

- **Onset:** 2–4 h. **Duration:** up to 24 h.
- Metabolised in liver to active metabolites excreted renally → accumulate in renal failure.
- **t½:** 10 h.

PHARMACODYNAMICS

- Binds **SUR1 on pancreatic β-cell KATP channels** → closes them → depolarisation → Ca^{2+} influx → ↑ insulin release (requires functioning β-cells).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Weight gain.
- Nausea, epigastric discomfort.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Severe, prolonged hypoglycaemia (may recur for 24–72 h; requires admission and IV glucose).
- Cholestatic jaundice; blood dyscrasias (rare).
- SIADH (hyponatraemia).
- Haemolysis in G6PD deficiency.

CONTRAINDICATIONS & CAUTIONS

- Type 1 diabetes; DKA.
- Severe renal or hepatic impairment.
- Elderly (relative); porphyria; sulfonamide hypersensitivity.
- Breastfeeding.

KEY INTERACTIONS

- ↑ **hypoglycaemia:** fluconazole, miconazole, co-trimoxazole, clarithromycin, alcohol, β-blockers (mask), warfarin.
- ↓ **effect:** steroids, thiazides, rifampicin.

NURSING MANAGEMENT

- Check glucose regularly; give with breakfast.
- Hold if patient not eating; seek advice.
- Hypoglycaemia: treat and monitor for at least 24–72 h (rebound).
- Monitor renal function, HbA1c, weight.

PATIENT EDUCATION

- Take with breakfast; never skip meals.
- Recognise and treat hypos; carry glucose.
- Avoid alcohol.
- Tell doctors about new antibiotics or antifungals.

KEY POINTS

- Long-acting sulfonylurea — ↑ insulin release.
- Prolonged hypoglycaemia; avoid in elderly/renal.
- Max 20 mg/day (US label).
- Weight gain.

SOURCES CHECKED [600] Glyburide, oral (US label) [601] Glyburide, oral (US label, Teva Pharmaceuticals USA)

Full references p. 432–457